Yifei Zhang

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Education

B.Sc. in Computer Science and Technology — **Nanjing University**
*Computer & Financial Engineering Elite Class, **Top-Notch Talent Program*** — *Sep 2020 – Jun 2024*

M.Sc. in FinTech and Financial Engineering — **Nanjing University**
School of Management & Engineering, supervised by Prof. Honghai Yu — *Sep 2024 – Jun 2027*

Internship Experience

Shanghai Xingyun Zhili Institute of Artificial Intelligence (XYZ AI Lab) — *2026.04 – 2026.07*
Research intern advised by Xiao Yang and Jiang Bian

- Worked on **Bounded AI4AI** for LLM post-training; led the development of the **evaluation harness** for agentic search in the open-source Agent Runtime **AxisAgentic** and designed **SFT data recipes**; XYZ-Aquila-mini achieved **SOTA among models of comparable size** on seven Deep Search benchmarks, with model weights, training data, and the runtime harness all open-sourced.

Microsoft Research Asia | Machine Learning Group — *2025.05 – 2026.01*
Research intern advised by Xiao Yang and Weiqing Liu

- Worked on MLE agents and LLM fine-tuning agents within the open-source **RD-Agent** framework; led the **Gome** algorithm design, cross-model scaling experiments, and the **FT-Dojo** benchmark; results published at **ACL Findings 2026** and **ICML 2026**.

The Chinese University of Hong Kong, Shenzhen | School of Data Science — *2024.06 – 2025.03*
Research intern advised by Prof. Benyou Wang

- Worked on financial-LLM evaluation and multi-agent social simulation; led the **UCFE Benchmark** and **Twin-Market** projects; results published at **NAACL Findings 2025** and **NeurIPS 2025**, and awarded **Best Paper** at the ICLR Financial AI Workshop (1/53).

Projects

Bounded AI4AI for LLM Post-Training — 2026.04 – 2026.07

- **Evaluation Harness for Agentic Search**: contributed to building and open-sourcing the Agent Runtime **AxisAgentic**, and led the development of a standardized evaluation harness across seven Deep Search benchmarks; under a controlled setup with fixed model, tools, and endpoint, systematically ablated context-management and verification strategies (retry, summary, self-verify, etc.) to quantify the **performance ceiling achievable by harness-only optimization**, lifting the average score across the seven benchmarks by over **20 points**.
- **Bounded AI4AI Paradigm & SFT Data Recipe Iteration**: contributed to building and validating a research paradigm in which **humans define goals, exploration boundaries, and acceptance gates, while agents perform automated exploration and accumulate experience**; automatically chained the four-stage loop Researcher → Developer → Reviewer → Recorder, covering data curation/synthesis, training/evaluation, and experience archiving, with support for parallel multi-experiment exploration and cross-reproduction; achieved unattended data-recipe iteration within preset gates, improving the model's accuracy on a private evaluation set by **8%**.
- The above work supported the open-source release of **XYZ-Aquila-mini (35B) and pro (397B)**: mini achieved **SOTA among models of comparable size** on seven Deep Search benchmarks, while pro attained the **highest publicly reported scores** on BrowseComp-ZH, LiveBrowseComp, and WideSearch; model weights, training data, and the runtime harness are all open-sourced.

FT-Dojo (RD-Agent 13,000+ Stars) | First Author — 2025.09 – 2026.01

- Built the **first interactive benchmark for autonomous LLM fine-tuning**, covering **13 tasks across 5 domains** (math, chemistry, finance, patents, and table QA), where the agent must autonomously complete the full loop of data curation / cleaning / synthesis → training → evaluation → iterative diagnosis.

- Systematically revealed three core bottlenecks of autonomous fine-tuning: context explosion, inefficient exploration due to missing fail-fast mechanisms, and misinterpretation of multi-level evaluation feedback; the proposed FT-Agent achieved the **best results on 10 of 13 tasks**, **outperforming frontier baselines such as Claude Code and Codex**.

Gome (RD-Agent 13,000+ Stars) | First Author — 2025.05 – 2025.09

- Observed that existing MLE agents, despite adopting different search topologies, process execution signals (error traces, training logs, code diffs) in the same way—**compressing them into scalar scores for candidate ranking, which discards the information of "which direction to improve"**. Inspired by gradient-based optimization, Gome shifts the role of feedback from ranking to **update**: it performs structured reasoning over execution signals to guide the direction of code modification (analogous to gradients), and introduces experiential memory to accelerate convergence (analogous to momentum) and multi-trajectory parallelism to broaden search coverage (analogous to distributed SGD).
- Achieved a **35.1% any-medal rate (open-source SOTA)** under the MLE-Bench closed-world protocol; compared against tree-search baselines across 10 models (GPT-4o-mini → GPT-5) and discovered a **crossover phenomenon**—tree search wins with weaker models while Gome prevails significantly with stronger ones (**+7.1% on GPT-5**)—revealing the distinct scaling behaviors of the two paradigms as model capability grows.

TwinMarket | First Author — 2024.10 – 2025.03

- Led the design and development of a large-scale LLM-driven multi-agent simulation platform for financial markets; each agent implements a perception–reasoning–action loop based on the BDI cognitive architecture, supporting scalable concurrent simulation of **1,000+ agents**.
- Designed the inter-agent social-network communication mechanism; without any explicit programming, the system spontaneously exhibited emergent collective phenomena such as **information cascades** and **herding effects**, reproducing stylized facts of financial markets including fat-tailed return distributions and volatility clustering.
- Delivered an invited oral presentation at the **ICLR 2025** Financial AI Workshop (Singapore) and received the **Best Paper Award** (1/53).

UCFE Benchmark | First Author — 2024.08 – 2025.01

- Designed and developed a user-centric evaluation framework for financial LLMs; constructed a dataset of 330 instances covering 17 task types based on a survey of **804 participants**.
- Adopted a GPT-4o-based user simulator together with the **LLM-as-Judge** paradigm, achieving strong alignment with human preferences (Pearson correlation **0.78**) and providing a reproducible, automated evaluation pipeline for assessing LLM capabilities.

AI-Trader | Lead of A-Share Module — 2025.11

- The first fully automated, real-time, contamination-free benchmark for LLM trading agents; led the construction of the A-share market module, implementing China-specific trading rules such as T+1 settlement and ±10% daily price limits, covering SSE 50 constituents.

Publications (* denotes equal contribution)

UCFE: A User-Centric Financial Expertise Benchmark for Large Language Models
Yuzhe Yang*, **Yifei Zhang***, Yan Hu*, Yilin Guo, Ruoli Gan, Yueru He, Mingcong Lei, Xiao Zhang, Haining Wang, Qianqian Xie, Jimin Huang, Honghai Yu, Benyou Wang.
NAACL Findings 2025 (CCF-B)

TwinMarket: A Scalable Behavioral and Social Simulation for Financial Markets
Yuzhe Yang*, **Yifei Zhang***, Minghao Wu*, Kaidi Zhang, Yunmiao Zhang, Honghai Yu, Yan Hu, Benyou Wang.
NeurIPS 2025 (CCF-A) & **Best Paper Award** at Financial AI @ ICLR 2025 (1/53).

Reasoning as Gradient: Scaling MLE Agents Beyond Tree Search
Yifei Zhang*, Xu Yang*, Xiao Yang*, Bowen Xian, Qizheng Li, Shikai Fang, Jingyuan Li, Jian Wang, Mingrui Xu, Weiqing Liu, Jiang Bian.
ACL Findings 2026 (CCF-A)

FT-Dojo: Towards Autonomous LLM Fine-Tuning with Language Agents
Qizheng Li*, **Yifei Zhang***, Xiao Yang*, Xu Yang*, Zhuo Wang, Weiqing Liu, Jiang Bian.
ICML 2026 (CCF-A)

Technical Reports

R&D-Agent: An LLM-Agent Framework Towards Autonomous Data Science
Technical Report for R&D-Agent Framework, 2025

AI4AI at Scale: A Full-Pipeline System for Enhancing LLM Agentic Capabilities
Technical Report for the new SOTA Deep Search Agent developed by XYZ AI Lab, 2026

Honors & Awards (Selected)

National Scholarship, Nanjing University (**top 0.2%**)	*2025.10*
National Gold Award, China International College Students' Innovation Competition	*2025.10*
Outstanding Graduate Student Pacesetter, Nanjing University	*2025.11*
Stars of Tomorrow Award for Outstanding Interns, Microsoft Research Asia	*2026.01*
Kweichow Moutai "Pillars of the Nation" Talent Program (**only 5 university-wide**)	*2026.05*